SOC HOME LAB PROJECT

Technical Documentation & Portfolio

Analyst: Amit
Role Target: SOC Analyst L1
Date: March 2026

Tools: Wazuh | TheHive | Kali Linux | Nmap | Mimikatz | Sysmon | Windows 10 | Ubuntu

1. Project Overview

This home lab simulates a real-world Security Operations Centre (SOC) environment using 4 virtual machines. A Kali Linux attacker launches real cyber attacks against a Windows 10 victim. Wazuh SIEM detects the attacks automatically and creates cases in TheHive for investigation. This demonstrates practical SOC Analyst L1 skills including threat detection, alert triage, and incident response.

1.1 Lab Machines

Machine	OS	Role	Tools & IP
Machine 1	Windows 10	**VICTIM**	Wazuh Agent + Sysmon — 192.168.1.x
Machine 2	Ubuntu	**DEFENDER**	Wazuh Manager + Dashboard — 192.168.1.x
Machine 3	Ubuntu	**SOC PLATFORM**	TheHive 5 Case Management — 192.168.1.x
Machine 4	Kali Linux	**ATTACKER**	Nmap, Hydra, Mimikatz — 192.168.1.x

2. Lab Architecture & Diagrams

2.1 Lab Architecture

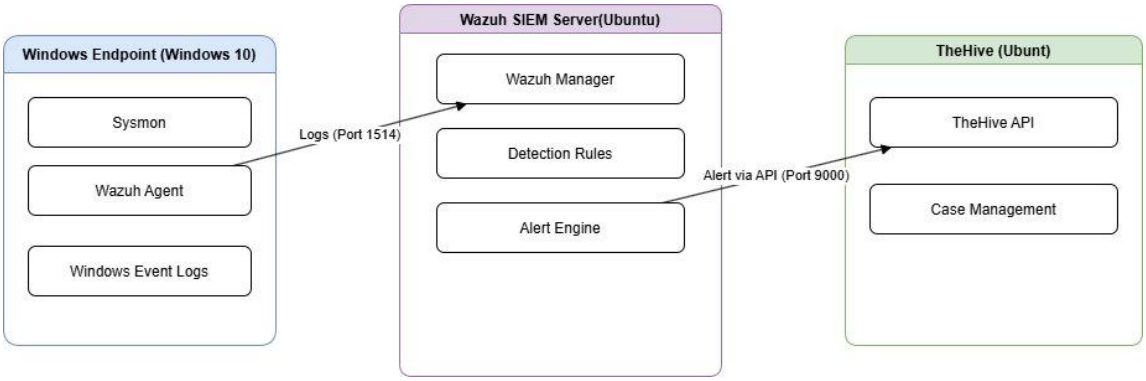


Figure 1 — SOC Lab Architecture: Windows Endpoint → Wazuh SIEM Server → TheHive

2.2 Event Flow

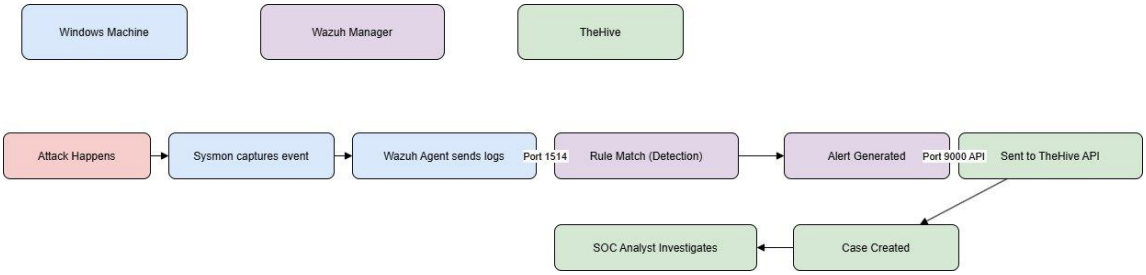


Figure 2 — Event Flow: Attack → Sysmon → Wazuh Agent → Rule Match → Alert → TheHive Case → SOC Analyst

3. Setup Documentation

3.1 Wazuh Manager — Machine 2

- Installed Wazuh full stack — Manager + Indexer + Dashboard
- Windows agent amit01 showing Active — 100% coverage

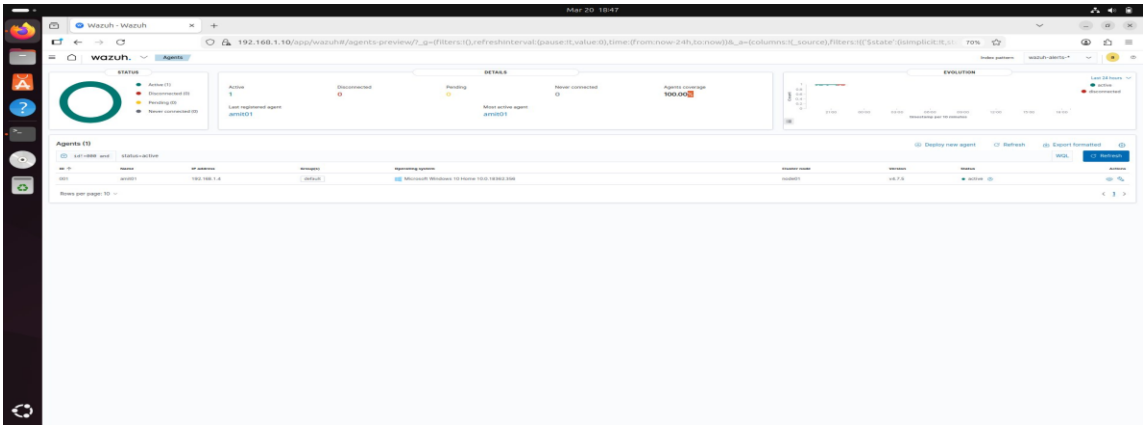


Figure 3 — Wazuh Agents dashboard: Windows agent amit01 Active on 192.168.1.4 — Microsoft Windows 10 Home

3.2 Wazuh Agent + Sysmon — Machine 1

- Wazuh Agent installed on Windows 10 Home
- Sysmon installed — generating detailed process, network and file events
- Sysmon events.

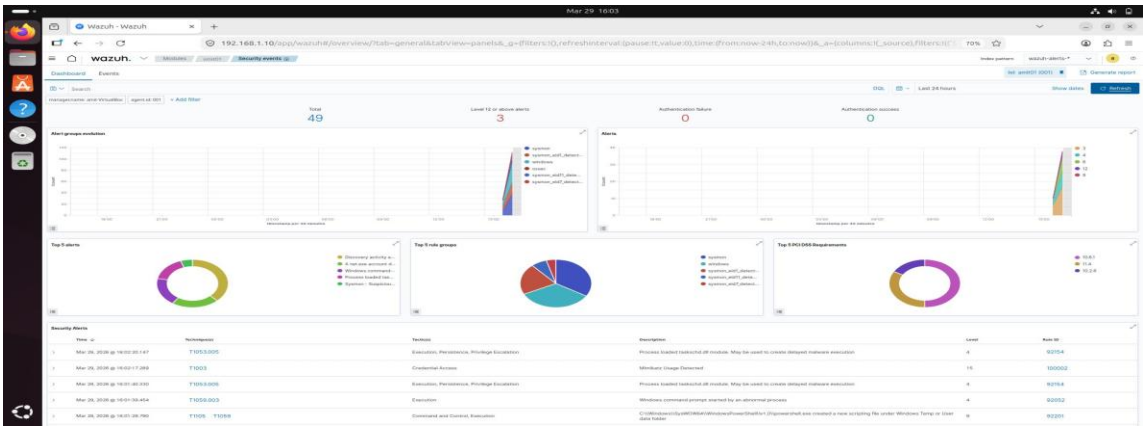


Figure 4 — Wazuh Security Events showing Sysmon detections including Mimikatz Rule 100002 CRITICAL T1003

3.3 TheHive — Machine 3

- TheHive installed with Community license from StrangeBee

- SOCLab organisation created — wazuh integration user configured
- Wazuh alerts automatically create alerts in TheHive via REST API

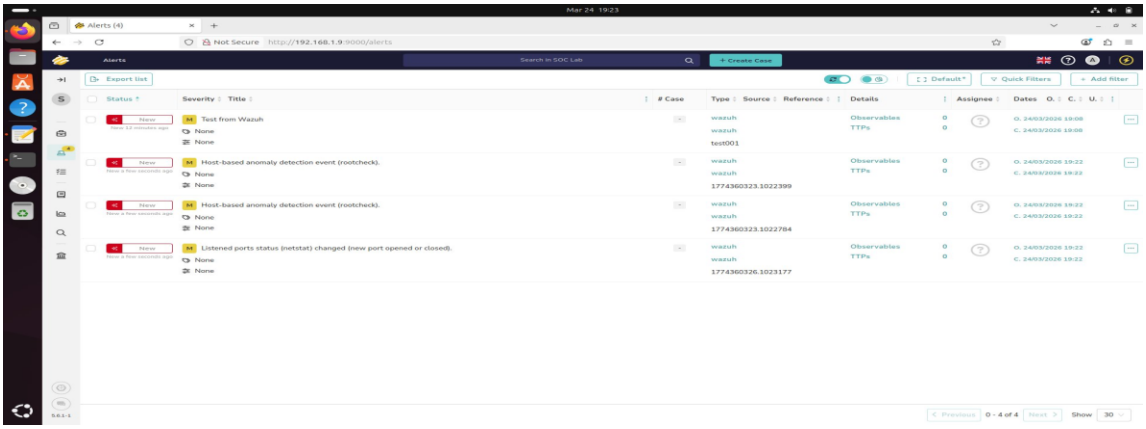


Figure 5 — TheHive Alerts showing initial connection from Wazuh — Test alert and host anomaly events

3.4 Kali Linux Attacker — Machine 4

- Kali Linux with all penetration testing tools pre-installed

Figure 6 — Kali Linux showing IP address 192.168.1.11 on eth0 Bridged Adapter

4. Attack Simulations

Three attack simulations were performed from Kali Linux and Windows. All attacks were detected by Wazuh and automatically created alerts in TheHive demonstrating a complete SOC detection and response workflow.

Attack 1 — Nmap Port Scan

MITRE: T1046 — Network Service Discovery | Severity: Low | Tool: Nmap

Field	Details
Attack Type	Network reconnaissance — port scanning
Attacker	Kali Linux — 192.168.1.x
Target	Windows 10 — 192.168.1.x
Command	nmap -sV -sC 192.168.1.x
Wazuh Detection	Multiple Firewall drop events
Alert Severity	Medium
Date	March, 2026

Kali — Nmap Running:

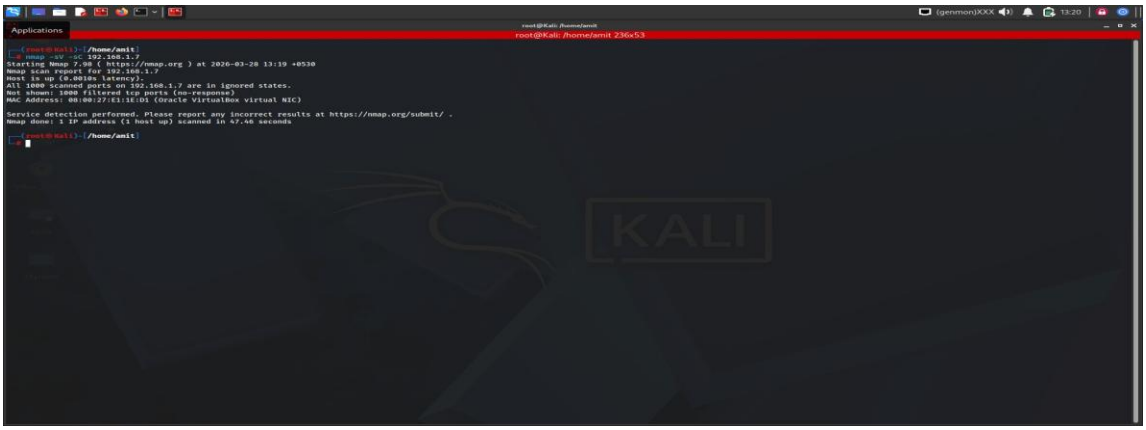


Figure 7 — Kali Linux running nmap -sV -sC against Windows target 192.168.1.7

Wazuh — Detection:

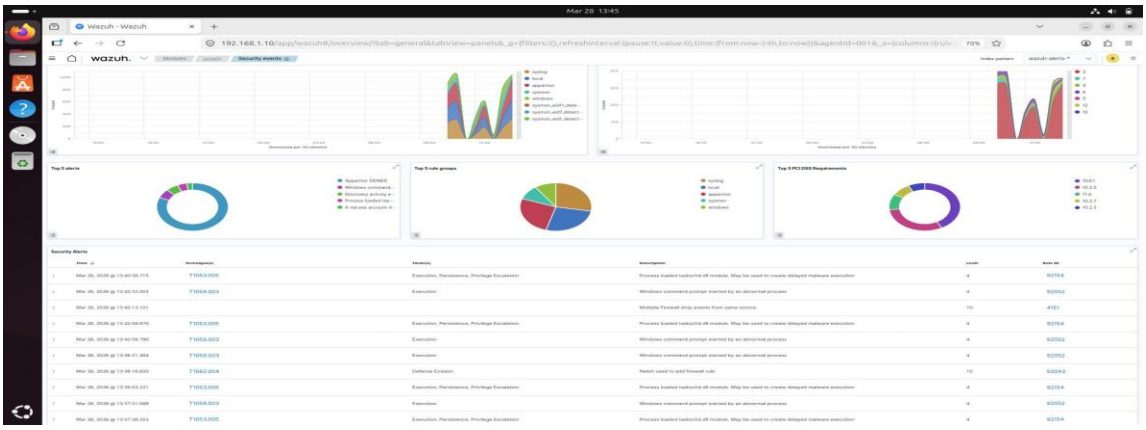


Figure 8 — Wazuh Security Events showing firewall events and MITRE technique detections triggered by Nmap scan

TheHive — Cases Created:

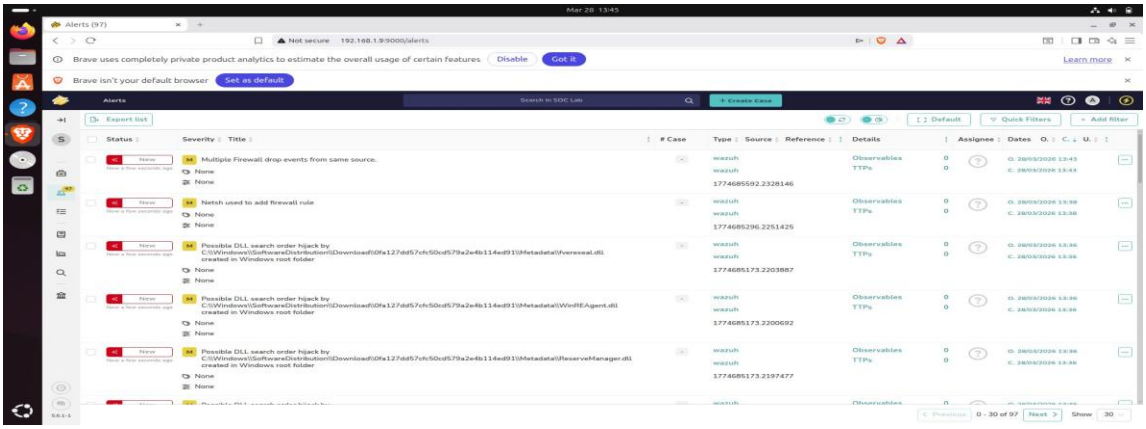


Figure 9 — TheHive showing 97 alerts including Multiple Firewall drop events and DLL hijack detections from Nmap scan

Attack 2 — Brute Force Login Attack

MITRE: T1110 — Brute Force | Severity: Medium | Tool: Manual + net use

Field	Details
Attack Type	Brute force — multiple failed login attempts
Attacker	Windows PowerShell — net use command
Target	Windows 10 local administrator account
Command	net use \\localhost\IPC$ /user:administrator wrongpassword
Wazuh Detection	Multiple authentication failures
Alert Severity	Medium
Date	March , 2026

Attack — Brute Force Running on Windows:

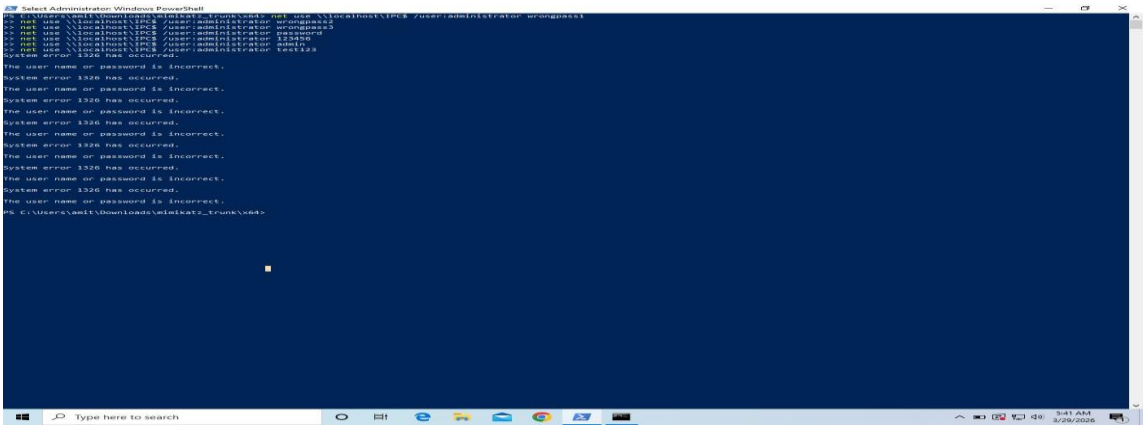


Figure 10 — Windows PowerShell showing manual brute force — 7 failed login attempts — System error 1326

Wazuh — Detection:

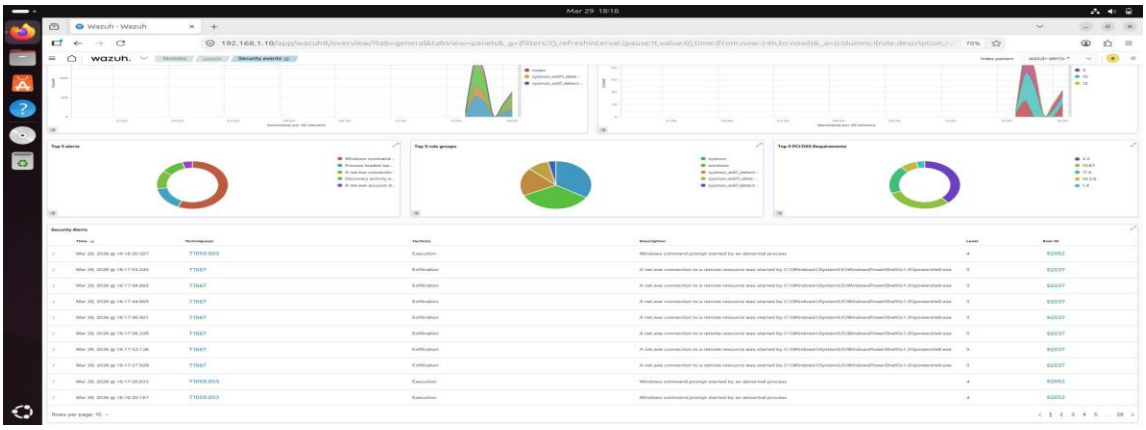


Figure 11 — Wazuh Security Events showing MITRE T1567 Exfiltration and T1059 Execution alerts triggered by attack activity

TheHive — Cases Created:

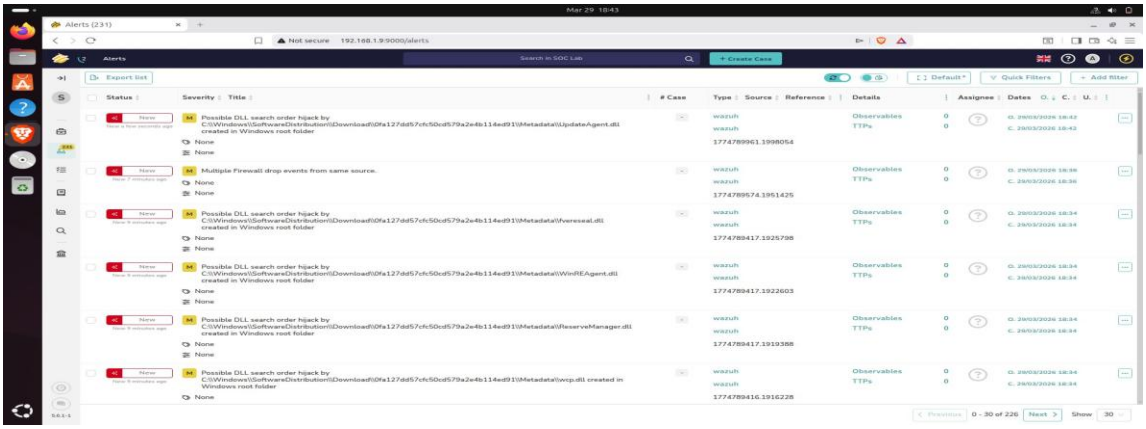


Figure 12 — TheHive showing 231 alerts including Multiple Firewall drop events and DLL search order hijack detections

Attack 3 — Mimikatz Credential Dumping

MITRE: T1003 — OS Credential Dumping | Severity: CRITICAL | Tool: Mimikatz 2.2.0

Field	Details
Attack Type	Credential dumping
Target	Windows 10 — 192.168.1.x
Tool	Mimikatz 2.2.0 x64 — downloaded and executed on Windows
Location	C:\Users\amit\Downloads\mimikatz_trunk\x64
Wazuh Rule	100002 — Mimikatz Usage Detected — Level 15 CRITICAL
Alert Severity	CRITICAL
Date	March , 2026

Attack — Mimikatz Running on Windows:

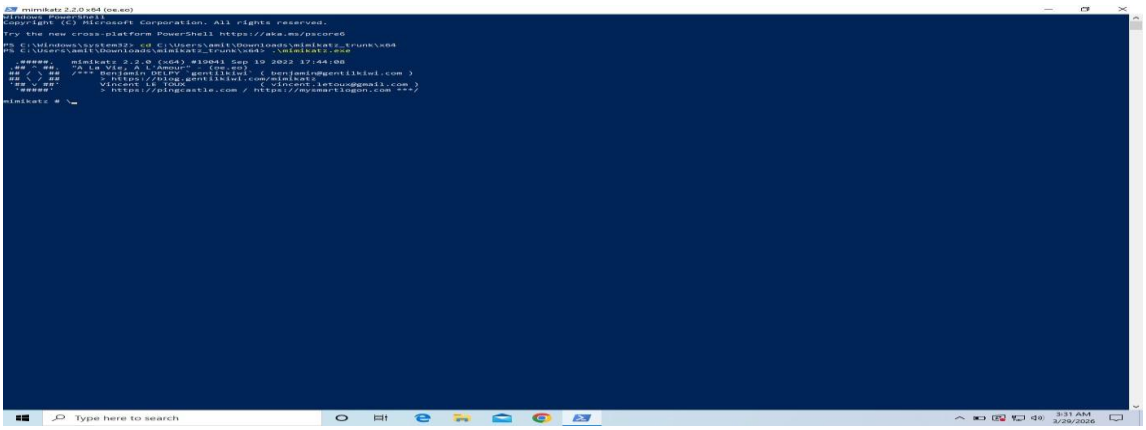


Figure 13 — Mimikatz 2.2.0 x64 running on Windows 10 PowerShell — executed from Downloads folder

Wazuh — CRITICAL Detection:

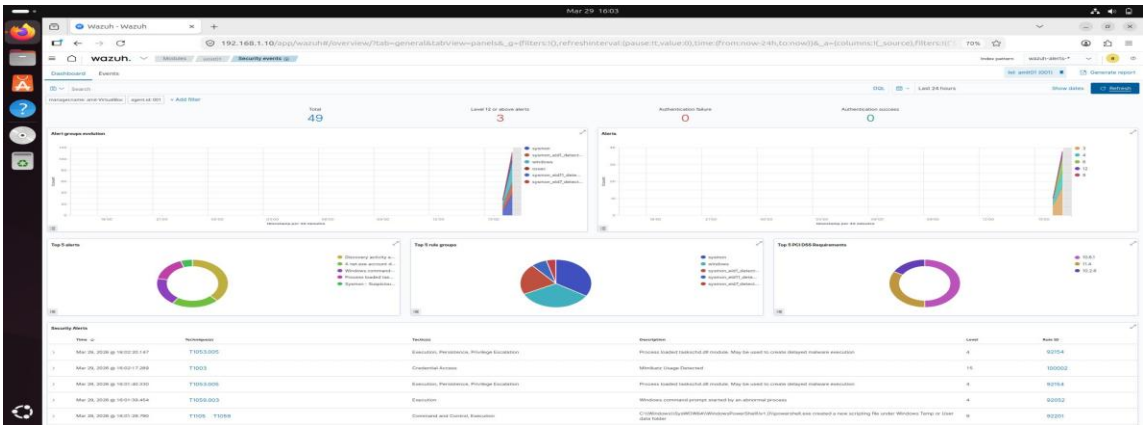


Figure 14 — Wazuh Dashboard showing Mimikatz Usage Detected (Rule 100002, Level 15 CRITICAL) — MITRE T1003 Credential Access

5. Incident Report — IR-2026-001

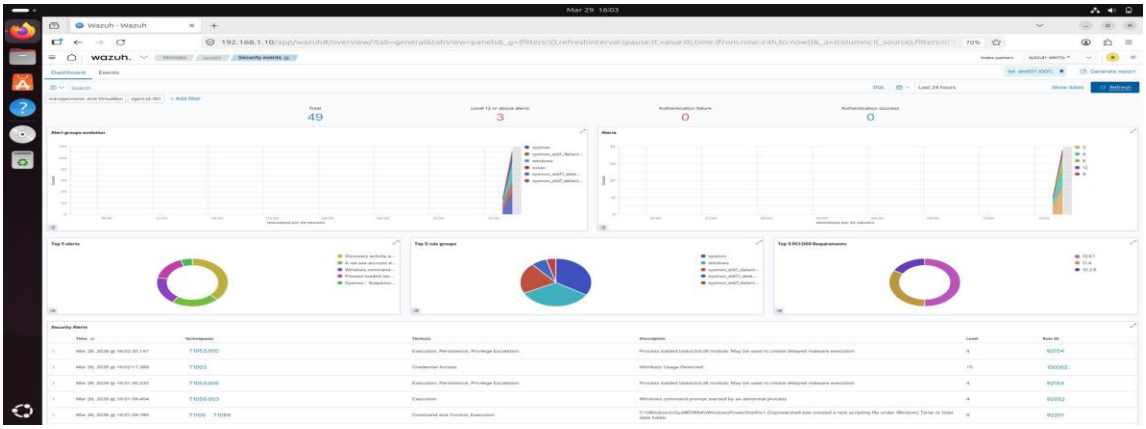
INCIDENT REPORT — IR-2026-001

Incident ID	IR-2026-001
Title	Mimikatz Credential Dumping on Windows 10 Endpoint
Date & Time	March 29, 2026 — 16:02
Severity	CRITICAL
Analyst	Amit
Victim	192.168.1.4 — Windows 10 — amit01
Wazuh Rule	100002 — Mimikatz Usage Detected — Level 15
MITRE ATT&CK	T1003 — OS Credential Dumping
Status	Resolved

5.1 Summary

Wazuh SIEM detected Mimikatz execution on Windows 10 endpoint (Machine 1 — 192.168.1.4) at 16:02 on March 29 2026. Sysmon logged LSASS process access which triggered Wazuh rule 100002 at severity level 15 (CRITICAL). This maps to MITRE ATT&CK T1003 — OS Credential Dumping. TheHive automatically created a case for analyst investigation.

5.2 Detection Evidence



Evidence — Wazuh showing Mimikatz Usage Detected Rule 100002 Level 15 CRITICAL with T1003 mapping

5.3 Investigation Steps

1. Received CRITICAL alert in Wazuh Dashboard — Rule 100002 Level 15
2. Identified MITRE technique T1003 — Credential Access — Credential Dumping
3. Opened corresponding case in TheHive for investigation
4. Reviewed Sysmon logs — confirmed LSASS process access by Mimikatz
5. Confirmed Mimikatz binary at C:\Users\amit\Downloads\mimikatz_trunk\x64
6. Verified attack was a controlled simulation — no actual credential exfiltration
7. Documented all findings and closed TheHive case as Resolved

5.4 Recommendations

- Enable Windows Credential Guard to protect LSASS memory from dumping
- Restrict SeDebugPrivilege — remove from non-administrator accounts
- Deploy EDR solution to detect and block Mimikatz signatures in real time
- Enable Protected Users security group in Active Directory
- Monitor Sysmon Event ID 10 for LSASS access attempts
- Implement application whitelisting to prevent unauthorised executables